Adithya A S

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SUMMARY

Software Engineer (2+ years) specializing in **end-to-end Full-Stack product development** (FastAPI, React.js, TypeScript). Expert in building and deploying **LLM-powered applications** utilizing **LangChain, OpenAI API, and LangGraph (React Agent)**. Proven ability to design and integrate **RAG pipelines** and **AI workflows** into production systems. Skilled in developing **scalable micro-frontends** for performance, and experienced with **cloud platforms (GCP, Azure)** and **Databricks** for data-driven application development. Seeking a **GenAI Engineer** role that leverages **AI innovation** with **Full-Stack engineering** to deliver impactful, high-quality solutions.

EXPERIENCE

Software Engineer

OPTUM **September 2023 - Present, Hyderabad, India**

- Led end-to-end development of enterprise-grade AI-powered web solutions, owning the Full-Stack architecture (backend, frontend) and CI/CD deployment pipelines across multi-cloud environments (AWS, GCP).
- Architected and deployed a hybrid RAG-based chatbot leveraging LangChain, LangGraph (React Agent), and OpenAI APIs. Implemented custom chunking strategies, MongoDB Atlas Vector Search, and semantic search workflows, improving response accuracy by 40%.
- Migrated legacy data ingestion pipelines to Azure Databricks, integrating automated data transformations for embedding generation and simultaneously pioneering optimized search infrastructure using MongoDB Atlas, reducing infrastructure costs by over 45% and lowering pipeline costs by 60%.
- Developed scalable micro-frontends using React.js and TypeScript, enhancing code reusability and optimizing application load times, while managing production systems to sustain 99.9% uptime with near-zero downtime deployments.
- Deployed and managed core cloud-native systems on AWS and GCP (Lambda, SageMaker, App Runner, API Gateway, S3, ECR, GCR, Cloud Run, and Compute Engine) to support scalable AI and data services.
- Devised robust CI/CD pipelines utilizing GitHub Actions and Terraform, streamlining automated testing, secure infrastructure provisioning, and multi-cloud deployments for enhanced reliability and accelerated release cycles.

AI FullStack Engineer

Flurf.AI - Personal Project **January 2025 - Present**

- Engineered an intelligent agent to automate complex UI workflows on documentation-heavy platforms (e.g., Postman) using natural language commands, reducing manual execution time by 70%.
- Implemented a full-stack Generative AI (GenAI) pipeline utilizing LangChain, OpenAI APIs, and MongoDB Atlas Vector Search for data processing (crawling, chunking, embedding) and semantic retrieval, achieving 90%+ query accuracy.
- Designed and deployed a hybrid RAG (Retrieval-Augmented Generation) architecture that interprets user intent, retrieves contextual data, and automates actions via a custom browser engine, resulting in 50% faster workflow completion.
- Built and deployed secure, scalable full-stack components including React, JWT authentication, PostgreSQL, Redis caching, and CI/CD pipelines, ensuring 99% system uptime and maintainability.
- Introduced performance optimizations including semantic caching and ETL enhancements, which improved query response times by 60% and reduced redundant computation by 45%.
- Documented technical specifications, APIs, and system integrations to facilitate cross-team collaboration, improving onboarding efficiency by 40% and ensuring software lifecycle maintainability.

EDUCATION

B.Tech in Computer and Communication Engineering

Amrita Vishwa Vidyapeetham • Coimbatore, India • 2023 • 7.8

CERTIFICATIONS

AWS Certified Cloud Practitioner

AWS Training and Certification • 2022

- Applied AWS best practices learned with this in real-world projects to design scalable, serverless architectures (Lambda, SageMaker) and automate deployments (ECR, CI/CD), improving efficiency and reducing infrastructure costs.

SKILLS

Generative AI & Data Engineering: Generative AI, Prompt Engineering, RAG pipelines, LLM integration, MLOps, Natural Language Processing (NLP), Semantic Search, Embeddings, Chunking, Vector Databases, Data Engineering, ETL, Data Pipelines, Cloud Computing

Programming & Frameworks: Python, FastAPI, LangChain, LangGraph, OpenAI API, Pydantic, React.js, TypeScript, Micro-frontends, Material UI, Crawl4Al, Vespa, MongoDB Atlas, Weaviate, Pinecone, JWT.

Cloud & DevOps: AWS (Lambda, SageMaker, API Gateway, S3, ECR), GCP (Cloud Run, Compute Engine, GCR), Azure Databricks, Docker, Kubernetes, CI/CD, Git, GitHub Actions, Jenkins, Argo Workflows / Airflow.

Tools & Methodologies: Agile Scrum, SDLC, Version Control, Semantic Caching, Performance Optimization, Problem-Solving, Team Collaboration, Communication.